

Dira Amanda Sabrina

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Highlights

- **ML Engineering:** Built and deployed production ML pipelines using Python, AWS SageMaker, and OML4Py for banking and manufacturing clients.
- **Generative AI & LLMs:** Built agentic AI systems and LLM-powered applications using Gemini API, LangChain, and technical exposure to RAG (Pinecone) and Transformers (HuggingFace library).
- **Data & Business Intelligence:** 4+ years delivering 10+ enterprise dashboards for FMCG, banking, manufacturing, and telco clients using Power BI, SQL, and Excel.
- **Advanced Study:** Master of Data Science at the University of Sydney with Australia Awards Scholarship Indonesia. Coursework in deep learning, NLP, supervised/unsupervised ML, and cloud computing (AWS).
- **Stack:** Python, SQL, PyTorch, AWS cloud (EC2, S3, RDS, SageMaker), HuggingFace, LangChain, Power BI, Git

Experience

Industry Experience Program (Sydney, Australia), Project Leader

November 2025 – December 2025

- Led a team of 5 to deliver a digital marketing strategy for an Australian health-tech startup, covering online visibility, competitor and market analysis, brand positioning, and partnership opportunities.
- Managed end-to-end project delivery within a two-week timeline, coordinating team workflow, tracking milestones, and aligning output to client feedback. Deliverable: measurable recommendations to address the startup's challenges.

Metrodata (Jakarta, Indonesia), Machine Learning Engineer

August 2024 – May 2025

- Improved profit prediction accuracy to 85% for a banking client by migrating and redeploying a regression model to an on-premise Oracle environment.
- Developed and deployed an end-to-end classification pipeline on AWS SageMaker to detect defective spare parts from imbalanced manufacturing data, reducing manual QC process effort with an automated system.
- Collaborated with the data engineer team to design and validate data pipelines, ensuring clean and consistent training data for ML projects in banking and manufacturing.

Metrodata (Jakarta, Indonesia), Business Intelligence Consultant

March 2021 – July 2024

- Delivered 15+ Power BI dashboards across FMCG, banking, manufacturing, and telco clients
- Employed Excel and SQL to analyse client datasets, developing data models, KPIs, and dashboards that provided high-level business insights.
- Collaborated with data engineers, project managers, and other cross-functional teams to maintain data quality, consistency, and reliable output for clients.

Projects

LLM-Powered Document Summariser

University project, Cloud Computing assignment

- Exposure to full AI product deployment lifecycle: building a document summarisation web app integrating the Gemini LLM API for content extraction, deployed on AWS infrastructure.
- Designed and configured AWS backend services (network, storage, database, API gateway) to support a production-oriented LLM application.
- Stack: Python, Gemini API, AWS

Agentic AI Marketing Decision System for Bank Deposit Prediction

Personal Project

- Built a simple end-to-end agentic AI system combining a supervised ML classifier, a rule-based decision agent, and an LLM explainer to support bank marketing teams on term deposit customer outreach.

- Designed a three-layer architecture: prediction, decision policy (rule-based) and LLM-generated explanation
- Integrated Gemini via LangChain to generate concise and actionable marketing guidance from agent outputs to bridge ML predictions to non-technical users
- Stack: Python, scikit-learn, Gemini API, LangChain, Streamlit, Git version control
(github.com/amandasbrn/bank-deposit-classif-agent)

AI Meal Planner

Personal Project

- Built a personalised recipe recommendation app that accepts ingredient inputs, calls Gemini API with Google Search grounding to suggest food recipes, and saves recipes to a local SQLite3 database
- Stack: Python, Gemini API, SQL (github.com/amandasbrn/ai-meal-planner)

Multi-Task LSTM for Text-to-SQL Generation

University project, Natural Language Processing assignment

- Built a multi-task LSTM model in PyTorch that simultaneously predicts NER tags per word and SQL query templates from user question: implementing a custom dataset class, full training loop with dev-set validation.
- Stack: Python, PyTorch, LSTM, NER, Sequence modelling
(github.com/amandasbrn/multitask-lstm-nlp-assignment3)

Bankruptcy Early-Warning System for Australian Charities

University project, Statistics assignment

- Built an ML classification model using 2023–24 financial data from the Australian Charities and Not-for-Profits Commission (ACNC) to detect early solvency risk indicators across non-profit organisations.
- Stack: Python, scikit-learn, pandas, classification with XGBoost / Random Forest

Education

The University of Sydney, Master of Data Science

August 2025 - present

- Relevant coursework: Natural Language Processing (RAG architecture with Pinecone, HuggingFace, Transformers-based model; self-attention, encoder and decoder, LSTM, text-to-SQL, data annotation, AI agent), Deep Learning (PyTorch, neural network architecture, multimodal), Cloud Computing (AWS), Supervised/Unsupervised Machine Learning. *See Projects section for applied work.*

Universitas Tarumanagara, Bachelor of Computer Science

2021

- Specialized in Business Intelligence; coursework: SQL, data mining, statistics, database, data warehouse.
- Best Thesis Award 1st Place: Air Pollution Monitoring and Forecasting Dashboard: designed an end-to-end BI solution covering ETL, data warehouse architecture, forecasting models in R, and a Power BI front-end. Selected as best thesis in the Information System undergraduate program.

Achievement

Australia Awards Scholarship

2025

- Awarded by DFAT to pursue a Master of Data Science at the University of Sydney, Australia: a highly competitive merit-based scholarship for Indonesian students

Best Employee of Metrodata on Quarter 3

2023

- Recognised for delivering high-quality client projects that led to contract extensions, generating additional revenue for the data analytics department.

Volunteer

DQLab, Artificial Intelligence Bootcamp Mentor Assistant

January 2023 - March 2023

- Assisted lead mentors in class, facilitated discussion and code reviews. Topics; data preparation, feature engineering, machine learning modeling, neural network, tuning, and deployment.